Healthy Skepticism · Session 1 take-home summary

Why this course exists

Medical claims reach you already packaged. Someone picked the number and wrote the headline, and those choices tend to run one way.

The aim is to empower you to have informed discussions with your doctor and make better health decisions. Healthy skepticism sits between two failures: blind trust on one side, cynicism on the other. The goal is to ask hard questions without dismissing good science.

We do the math so you do not have to. Figures, not formulas, with a lot of real-world examples, a companion website, this take-home paper each day, and deeper whitepapers on single topics for anyone who wants them. Every assertion is backed up.

Everything in this course is done by counting. No formulas and no bell curves: whole people, out of a stated number of people. If a claim can't be put into counted squares, that tells you something about the claim.

The 100-square diagram used in class is illustrative, chosen to make the counting visible rather than to describe any particular study.

The five faults

The course is built around five recurring faults in how medical evidence is reported and read.

Fault	The question it raises	Session
1. Relative reporting	What does a 36% improvement from taking a drug really mean?	Day 1
2. Probability reversal	What does a positive COVID test really mean?	Day 2
3. Arbitrary categories	How concerned should you be if your systolic BP goes from 119 to 120?	Day 3
4. The Bernoulli fallacy	Why does the published medical advice keep changing?	Day 3
5. The causal leap	Does bird watching prevent dementia?	Day 4

Today is Fault 1. It comes first because it is the most common, and you will meet it again in an advertisement this week.

Addressing uncertainty

Medicine is uncertain. Every claim is a claim about chances. A drug doesn't prevent your heart attack; it changes how often heart attacks happen in a group of people like you. The honest way to state that is as a count.

So the working question throughout is not "does this work?" but "out of how many, how many?"

Fault 1: relative reporting

A well-known statin advertisement led with a 36% reduction in heart attacks. The number is not fabricated. It is a real result from a real trial. The problem is that 36% is a ratio between two other numbers, and neither of them appears in the headline.

The trial figures, expressed the way this course expresses everything, look like this.

Out of 1,000 people like those in the trial	Heart attacks
Taking the placebo	about 30
Taking the drug	about 19
Difference	about 11 per 1,000

Eleven fewer events per thousand people. Divide 11 by 30 and you get the 36%. The headline number is the ratio; the eleven is the benefit.

> **The honest sentence:**
> Of 1,000 people like those in the trial, about 30 had a heart attack without the drug and about 19 with it. Eleven avoided one. About 91 people had to take the drug for one of them to benefit.

That last figure has a name: the **number needed to treat**, here 1,000 divided by 11, about 91. It is the most useful number in the report, and it almost never makes the headline.

Advertised claim based on ASCOT-LLA (Sever PS et al., Lancet 2003;361:1149-1158); event counts rounded to whole people per 1,000 for teaching.

Where the missing piece hides

The baseline risk isn't hidden outright; it's in the fine print and in the trial paper. The ad separates the ratio from the number it's a ratio of, and prints only the ratio.

This matters most when the baseline is small. A 50% reduction in a risk of 2 in 10,000 removes 1 case in 10,000. The same 50% applied to a risk of 2 in 10 removes 1 case in 10. The headline is the same; the number of people helped differs a thousandfold.

The three questions to ask

1. **36% of what?** A percentage with no stated baseline is not yet a fact.
2. **What was the risk without the treatment?** This is the number the ratio was computed from.
3. **Out of how many people, how many actually benefited?** Ask for it as whole people out of a thousand.

If a claim survives all three, it's probably reported fairly. If any of the answers isn't available, be suspicious.

What to ask, and what comes next

Bring the third question to your next appointment. It's a fair question, and it has an answer: "If a thousand people like me took this, how many would be helped, and how many would have had the problem anyway?"

Next session: probability reversal. A test is measured one way round and read the other way round. What a positive result means to you is not what the test's advertised accuracy measures, and the gap between them can be very large.

Further reading

- Aubrey Clayton, *Bernoulli's Fallacy* (Columbia University Press, 2021). Where the standard methods came from and what they quietly assume.
- Judea Pearl and Dana Mackenzie, *The Book of Why* (Basic Books, 2018). How causal questions differ from questions about correlation.

Course materials and contact

Slides, videos and the whitepaper library are at healthy-skepticism.com. Decks live in the student area, behind the password given out in class.

Murray Cantor PhD, murray@murraycantor.com

Joel Keenan MD, drkeenan@mdvip.com